Hidden Breast Cancer Disparities in Asian Women: Disaggregating Incidence Rates by Ethnicity and Migrant Status

Scarlett Lin Gomez, PhD, Thu Quach, PhD, Pamela L. Horn-Ross, PhD, Jane T. Pham, MPH, Myles Cockburn, PhD, Ellen T. Chang, PhD, Theresa H.M. Keegan, PhD, Sally L. Glaser, PhD, and Christina A. Clarke, PhD

Objectives. We estimated trends in breast cancer incidence rates for specific Asian populations in California to determine if disparities exist by immigrant status and age.

Methods. To calculate rates by ethnicity and immigrant status, we obtained data for 1998 through 2004 cancer diagnoses from the California Cancer Registry and imputed immigrant status from Social Security Numbers for the 26% of cases with missing birthplace information. Population estimates were obtained from the 1990 and 2000 US Censuses.

Results. Breast cancer rates were higher among US- than among foreign-born Chinese (incidence rate ratio [IRR] = 1.84; 95% confidence interval [CI] = 1.72, 1.96) and Filipina women (IRR = 1.32; 95% CI = 1.20, 1.44), but similar between US- and foreign-born Japanese women. US-born Chinese and Filipina women who were younger than 55 years had higher rates than did White women of the same age. Rates increased over time in most groups, as high as 4% per year among foreign-born Korean and US-born Filipina women. From 2000–2004, the rate among US-born Filipina women exceeded that of White women.

Conclusions. These findings challenge the notion that breast cancer rates are uniformly low across Asians and therefore suggest a need for increased awareness, targeted cancer control, and research to better understand underlying factors. (*Am J Public Health.* 2010;100:S125–S131. doi:10.2105/AJPH.2009.163931)

Breast cancer is the most common cancer diagnosed among US women.[1] Nevertheless, in the United States, incidence rates vary substantially by standard racial/ethnic categories, with the lowest rates reported for Asians as a single group.[2] However, within this group, breast cancer incidence rates vary further by specific ethnicity,[2,3] with a nearly 3-fold difference between populations with the highest rate (Japanese women: 126 per 100000) and the lowest rate (Laotian women: 44 per 100000).[2] Moreover, within each of the Asian populations, there is substantial heterogeneity in lifestyles, health care practices, and risk factors[4–14] that may correspond to important differences in breast cancer rates.

Higher breast cancer incidence rates among Asian women living in the United States than among those living in Asian countries,[15] together with elevated breast cancer risks associated with immigration to and longer residence in the United States,[3,16,17] have been centerpiece evidence for major roles for environmental, nongenetic factors in breast cancer causation. Earlier research has shown that immigrant status (US-born vs foreign-born) is associated with breast cancer risk through changes in reproductive factors (e.g., higher age at first live birth, lower breast feeding rates, earlier onset of menstruation)[18,19] and lifestyle factors (e.g., diet)[3] but could also indicate variations in other environmental exposures.[20]

Examination of breast cancer incidence rates by immigrant status could inform these and other issues in breast cancer etiology. However, calculation of the accurate incidence rate trends by immigration status has been hampered by (1) nonrandom missing data on immigrant status for approximately 30% of cancer registry cases[21–23] and (2) the lack of annual US Census estimates of Asian population counts according to immigrant status. To overcome these limitations, we augmented data on cancer cases from the population-based California Cancer Registry (CCR) through statistical imputation of immigrant status for cancer cases with missing birthplace and then used robust demographic methods to compute corresponding population estimates. With the resulting enhanced data resource, we sought to estimate breast cancer incidence rates by immigrant status, age, and time for specific Asian populations in California, the US state with the largest and most diverse Asian population.[24]

METHODS

We obtained information from the CCR regarding all females residing in California who were diagnosed with a primary invasive breast cancer (*International Classification of Disease for Oncology, 3rd Edition,* [*ICD-O-3*] site codes C50.0–50.9) from January 1, 1988, through December 31, 2004. The CCR comprises 3 of the National Cancer Institute's Surveillance, Epidemiology, End Results (SEER) program registries and collects information on patient race/ethnicity and birthplace from hospital medical records, which is obtained primarily through self-report; by assumption of hospital personnel; on inference from other information including race/ethnicity of parents, maiden name, surname, and birthplace; and from death records.[25]

In this study, we included 21147 breast cancer cases among the 6 Asian ethnic populations that together comprised 91% of all Asian patients with breast cancer in the CCR during the study period. Of these, 5732 (27.1%) cases were among Chinese women, 3888 (18.4%) were among Japanese women, 7583 (35.9%) were among Filipina women, 1304 (6.2%) were among Korean women, 1130 (5.3%) were among South Asian women (including Asian Indians, Pakistanis, Sri Lankans, and Bangladeshis), and 1510 (7.1%) were among Vietnamese women. We excluded from further consideration cancer patients classified by the registry as "Other Asian or Asian, not otherwise specified" (n=1099).

Our prior research showed that patients in the cancer registry with unknown birthplace data were more likely to be US-born than those with available data.[21–23] Thus, we expect that rates based on random imputation of immigrant status would be underestimated for the US born and overestimated for the foreign born. To more accurately classify patient immigrant status, we used 2 sources of information about birthplace: (1) registry-based data available for 15 387 patients, or 72.8% of cases (69.7% from hospital medical records and 3.1% from death certificates) and (2) statistical imputation of immigrant status using the first 5 digits of the patient's Social Security Number (SSN), indicating the state and year of issuance for those who had unknown birthplace (n=5409, or 25.6% of patients).[26,27] With this approach, we imputed immigrant status as follows: individuals who received their SSN before age 25 years were considered US born, and those who received their SSN at or after age 25 years were considered foreign born. The age cut-point was set at 25 years based on self-reported nativity in previously interviewed cancer patients (n=1836)[22] and maximization of the area under the receiver-operating characteristic curve. The optimal positive predictive value of this single age cut-point was confirmed by using a logistic regression model with age at SSN issue as a continuous predictor of foreign-born status. This age cut-point resulted in immigrant status classifications associated with 84% sensitivity and 80% specificity for detecting foreign-born (i.e., 80% sensitivity for detecting US-born individuals), and was similar across Asian populations. The 351 individuals (1.7%) with missing or invalid SSNs were assigned an immigrant status based on the ethnicity–sex–age nativity distribution of the overall sample.

Population Data for Rate Denominators

From the 1990 and 2000 Census Summary File 3 (SF-3),[28,29] we obtained population counts by gender, race/ethnicity, immigrant status, and 5-year age group for the state of California. Data from the 20% Integrated Public-Use Microdata Sample of the Census also were used to estimate age- and immigrant status–specific population counts for the 6 Asian groups[30–32] by smoothing the distribution with a spline-based function.[33] For intercensal years, we estimated the percentage of foreign-born individuals by using cohort component interpolation and extrapolation methods, adjusting estimates to the populations by age and year provided by the California Department of Finance for years 1988 to 1989 and by the US Census for years 1990 to 2004 because of data availability. Incidence rates for US-born Koreans, South Asians, and Vietnamese were not stable enough to report because of 2 consequences of the recent immigration of these groups. First, the US-born populations were significantly smaller than the foreign-born populations, and, second, US-born populations have considerably younger age distributions, reducing stability of age-adjusted rates for cancers, which predominantly occur at older ages.

Statistical Analysis

We used SEER*Stat software[34] to compute age-adjusted incidence rates (standardized to the US standard million population for 2000) and 95% confidence intervals (CIs) among women. We compared incidence rates between US- and foreign-born Asian populations with incidence rate ratios (IRRs). We examined time trends between 1988 and 2004 using Joinpoint Regression software[35] to calculate the annual percentage changes (APC). We calculated APCs by fitting a series of least squares regression lines to the natural logarithm of the age-adjusted incidence rates, with calendar year as a regression variable.[35]

RESULTS

During the period 1988–2004, invasive breast cancer incidence rates were lower for all of the Asian ethnic groups than for non-Hispanic White women (Table 1). Table 1, which compares rates across ethnicity and

TABLE 1—Age-Adjusted Incidence Rates of Invasive Breast Cancer Among Women, by Race/Ethnicity and Immigrant Status: California Cancer Registry, 1988–2004

Race/Ethnicity	No. of Cases	Population, No.	Incidence Rate (95% CI)	Rate Ratio[a] (95% CI)
Non-Hispanic Whites	248 421	142 863 244	146.1 (145.5, 146.7)	
Asians (Aggregated)[b]	21 147	27 237 581	82.7 (81.6, 83.8)	
Foreign-born (Ref)	16 627	18 865 657	76.3 (75.1, 77.5)	1.00
US-born	4 520	8 371 924	120.6 (117.1, 124.3)	1.58 (1.53, 1.63)
		Asians (disaggregated)		
Chinese women	5 732	7 985 862	73.5 (71.6, 75.4)	
Foreign-born (Ref)	4 490	5 626 259	66.3 (64.4, 68.4)	1.00
US-born	1 242	2 359 603	122.1 (115.0, 129.0)	1.84 (1.72, 1.96)
Japanese	3 888	3 066 109	102.5 (99.3, 105.9)	
Foreign-born (Ref)	1 590	1 236 742	103.1 (97.8, 108.8)	1.00
US-born	2 298	1 829 367	106.1 (101.7, 111.0)	1.03 (0.96, 1.10)
Filipina	7 583	7 809 502	100.4 (98.1, 102.8)	
Foreign-born (Ref)	6 946	5 577 737	98.2 (95.9, 100.6)	1.00
US-born	637	2 231 765	129.5 (118.7, 141.0)	1.32 (1.20, 1.44)
Korean	1 304	2 914 486	46.3 (43.8, 49.0)	
Foreign-born (Ref)	1 212	2 306 719	44.5 (42.0, 47.2)	1.00
US-born	92	...	...	...
South Asian[c]	1 130	2 197 346	77.0 (72.1, 82.1)	
Foreign-born (Ref)	989	1 593 537	70.5 (65.7, 75.6)	1.00
US-born	141	...	...	...
Vietnamese	1 510	3 264 276	59.9 (56.7, 63.1)	
Foreign-born (Ref)	1 400	2 524 663	57.0 (53.9, 60.3)	1.00
US-born	110	...	...	...

Note. CI = confidence interval. Ellipses indicate that sample sizes were too small to report because of the potential for large errors associated with them. Breast cancer incidence rates are per 100 000 person-years, age-standardized to the 2000 US population.
[a]Ratio of the incidence rate in US-born to the incidence rate in foreign-born.
[b]Consists of the 6 Asian populations: Chinese, Japanese, Filipina, Korean, South Asian, and Vietnamese women.
[c]"South Asian" populations include Asian Indians, Pakistanis, Sri Lankans, and Bangladeshis.

immigrant status, shows that US-born Chinese and Filipina women had cancer rates that were approximately 80% and 30% higher, respectively, than were rates for their foreign-born counterparts. By contrast, rates were comparable for US-born and foreign-born Japanese women (106 and 103 per 100 000, respectively). Incidence rates were significantly higher for US-born Chinese (122 per 100 000; 95% CI=115, 129) and Filipina women (130 per 100 000; 95% CI=119, 141) than for US-born Japanese women.

We observed an increase in incidence rates across time periods in both the US-born and foreign-born aggregated Asian populations, although the increase appeared to be greater in the US-born Asian populations (Table 2, Figure 1). US-born Filipina and foreign-born Korean women had the largest increases in breast cancer incidence (4% yearly) over time. The annual increase in incidence rates was statistically significant for all groups except foreign-born Vietnamese and Japanese women. Rates in foreign-born Japanese women increased steadily from 1988 to 1999 before dropping substantially in 2001 to 2004. Across all time periods, the highest breast cancer incidence rate was observed for US-born Filipina women diagnosed from 2000 to 2004 (162.5 per 100 000; 95% CI=144.3, 182.2).

In age-specific data (Table 3), rates were higher among US-born than foreign-born groups across all age groups examined, except among Japanese women. Notably, incidence rates for US-born Chinese and Filipina women aged 44 years or older and those aged 45 to 54 years at diagnosis were higher than were corresponding rates for non-Hispanic White women, whereas incidence rates for women aged 55 years and older were considerably lower.

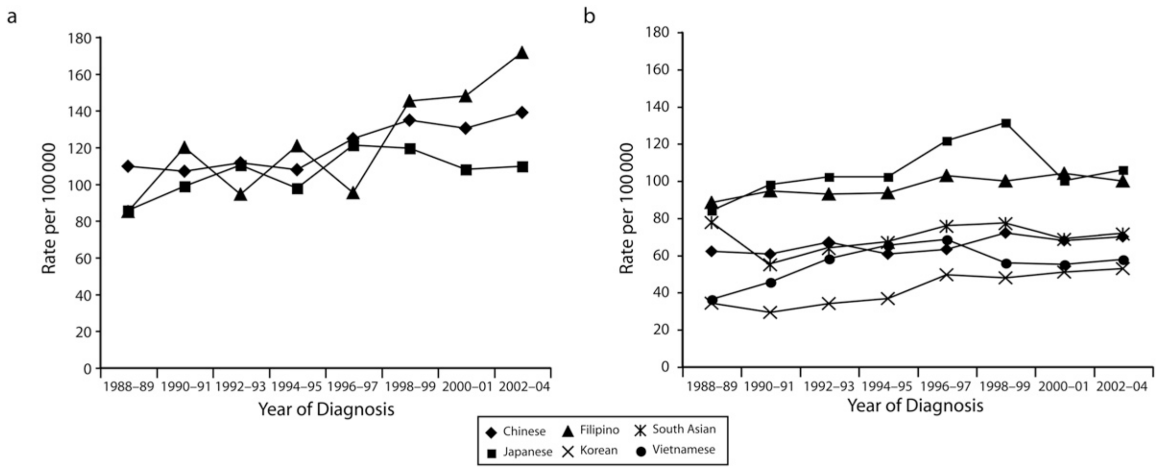
DISCUSSION

Using new, rigorously derived cancer and population data estimates for women by immigrant status, we identified subpopulations of US Asian women who appear to have higher burdens of breast cancer than were previously described. Specifically, we found that US-born Asian women have incidence rates of invasive breast cancer nearly 2-fold higher than do foreign-born Asian women in all groups examined except Japanese women. These findings suggested that breast cancer rates reported for specific US Asian populations are sensitive to the proportion of foreign-born women. Our results also suggest that rates for US-born Chinese, Japanese, and Filipina women approached and, among Filipina women exceeded, those for non-Hispanic White women; this pattern was consistent with a strong environmental influence on breast

TABLE 2—Age-Adjusted Incidence Rates of Invasive Breast Cancer Among Women, by Race/Ethnicity and Immigrant Status: California Cancer Registry, 1988–2004

	1988–1994			1995–1999			2000–2004			
Race/Ethnicity	No. of Cases	Population, No.	Incidence Rate (95% CI)	No. of Cases	Population, No.	Incidence Rate (95% CI)	No. of Cases	Population, No.	Incidence Rate (95% CI)	APC,% (95% CI)
Non-Hispanic Whites	96 695	59 971 664	143.0 (142.1, 143.9)	89 438	49 973 547	149.7 (148.7, 150.7)	76 235	41 363 602	145.6 (144.6, 146.7)	0.79 (0.5, 1.1); −4.8 (−7.5, −2.0)[a]
US-born Asians										
Aggregated[b]	1 322	2 778 015	104.1 (98.1, 110.4)	1 394	2 525 906	124.4 (117.8, 131.2)	1 804	3 068 003	135.9 (129.6, 142.4)	2.4 (1.5, 3.3)
Chinese	327	773 288	104.2 (92.2, 117.2)	392	714 587	128.8 (115.9, 142.6)	523	871 728	135.6 (124.0, 147.9)	2.0 (0.5, 3.4)
Japanese	783	741 176	99.3 (91.7, 107.3)	736	537 331	114.7 (106.2, 123.7)	779	550 860	108.7 (101.0, 117.0)	1.2 (−0.1, 2.6)
Filipina	143	732 649	100.5 (81.4, 123.6)	174	675 926	123.8 (102.5, 148.1)	320	823 190	162.5 (144.3, 182.2)	4.2 (2.2, 6.2)
Korean	20	...	...	23	...	...	49	...	...	...
South Asian[c]	22	...	...	44	...	...	75	...	...	...
Vietnamese	27	...	...	25	...	...	58	...	...	...
Foreign-born Asians										
Aggregated[b]	4 469	6 306 269	70.6 (68.4, 72.8)	5 300	5 810 196	79.3 (77.1, 81.5)	6 858	6 749 192	78.5 (76.6, 80.4)	1.1 (0.6, 1.6)
Chinese	1 237	1 879 295	62.8 (59.3, 66.6)	1 389	1 737 574	66.6 (63.1, 70.4)	1 864	2 009 390	68.9 (65.8, 72.2)	1.0 (0.3, 1.7)
Japanese	485	453 257	95.8 (85.6, 107.0)	536	370 528	122.0 (110.6, 134.4)	569	412 957	103.9 (95.1, 113.5)	1.2 (−0.5, 2.9)
Filipina	1 946	1 923 239	92.2 (88.0, 96.8)	2 178	1 713 057	99.9 (95.7, 104.4)	2 822	1 941 441	101.4 (97.7, 105.4)	0.8 (0.1, 1.5)
Korean	273	820 284	32.5 (28.5, 37.0)	392	704 934	46.8 (42.2, 52.1)	547	781 501	52.1 (47.8, 57.2)	4.2 (3.0, 5.5)
South Asian[c]	216	426 908	62.5 (53.4, 73.0)	332	499 412	76.3 (67.5, 86.2)	441	667 217	69.9 (62.9, 77.5)	...
Vietnamese	312	803 286	52.3 (46.2, 59.1)	473	784 691	61.6 (55.9, 68.1)	615	936 686	56.6 (52.0, 61.9)	2.4 (−0.2, 5.0)

Note. APC = annual percentage change; CI = confidence interval. Ellipses indicate that sample sizes were too small to report because of the potential for large errors associated with them. Breast cancer incidence rates are per 100 000 person-years, age-standardized to the 2000 US population.
[a]Joinpoint regression found multiple regression lines that fit the data. The first APC is for 1988-2001; the second is for 2001–2004.
[b]Consists of the 6 Asian populations: Chinese, Japanese, Filipina, Korean, South Asian, and Vietnamese women.
[c]"South Asian" populations include Asian Indians, Pakistanis, Sri Lankans, and Bangladeshis.



Note. Rates for Korean, South Asian, and Vietnamese women are not shown because of small denominations.

FIGURE 1—Age-adjusted breast cancer incident rates among California women who are (a) US born and (b) foreign born: California Cancer Registry, 1988–2004.

cancer risk. Our findings also suggested that US-born Asian women younger than 55 years in particular have higher risks of breast cancer than previously estimated. To our knowledge, our study is the first to suggest that rates among pre- and peri-menopausal US-born Chinese and Filipina women were higher than were those for non-Hispanic White women.

These relatively higher rates in Asian populations together with their younger age distributions lead to a considerably greater proportion of breast cancer cases diagnosed among young women than occurs in other racial/ethnic groups. For example, from 1988–2004, the percentage of breast cancers diagnosed among women younger than 55 years was 67.5% among US-born Filipina women but only 28.9% among non-Hispanic White women. Furthermore, age-specific incidence patterns among US- and foreign-born Asian women differed from patterns seen in non-Hispanic White women, with rates differing very little between women 55 years and older and those aged 45–54 years. By contrast, among non-Hispanic White women, incidence rates were nearly 2-fold higher in the 55 years and older age group. In this regard, the age-specific incidence patterns we estimated for US-born Chinese and Filipinas in particular were noteworthy in light of the age-specific incidence for African American women, which have been shown to be higher than that for non-Hispanic Whites among women diagnosed who were younger 40 years, but "crossing over" to be relatively lower for women diagnosed at older ages.[36,37] Although our 1988–2004 data show that the rates for women younger than 40 years were indeed higher for African American (15.1 per 100 000) than they were for non-Hispanic White women (12.8 per 100 000), the rates for US-born Chinese and Filipina women were even higher (20.3 and 19.8 per 100 000, respectively). Thus, the observed ethnic and immigrant differences in the age-specific patterns of breast cancer incidence relative to menopause may lend insights into breast cancer etiology among Asian women, particularly as these age-specific patterns relate to changes in acculturation that may be occurring during this critical time period.

Previous studies have reported comparable breast cancer incidence rates for US-born Japanese and non-Hispanic White women,[38,39] despite the lower prevalence among Japanese women of particular breast cancer risk factors (e.g., lower weight or body mass index and less alcohol consumption).[40] We identified increasing rates of breast cancer for US Japanese women, regardless of immigrant status, during the period 1988–2000, mirroring the rising rates of breast cancer in Japan,[41] which[42] nearly doubled between 1978 and 1998.[41,43,44] Women in Japan have become increasingly similar to US White women with respect to the population prevalence of lifestyle-related breast cancer risk factors. After World War II, the number of women in the Japanese workforce has risen,[45] which has produced a declining birthrate.[46,47] This societal shift may have resulted in relevant changes in the population distribution of breast cancer risk factors, such as higher socioeconomic status,[48] higher body mass index, fewer births, and later age at first birth.[42,49] Consequently, breast cancer risk profiles may be more similar for foreign- and US-born Japanese women than for other Asian populations, causing a narrowing in the incidence rate gap between Japanese immigrants to the United States and those who are US-born. In addition, it is likely that the foreign-born Japanese immigrated earlier in life, and thus may be more acculturated, than other foreign-born Asians, based on the younger age of SSN issuance among Japanese observed in our data.

Some of our results differ from those reported in the only prior examination of breast cancer incidence rates by immigrant status.[17] In an analysis based on 1973 to 1986 SEER data from Seattle, Washington, and San Francisco, Oakland, and Los Angeles, California, US-born Chinese and Japanese women had 50% higher incidence rates than did their foreign-born counterparts, whereas rates among US-born Filipinas were slightly lower than those among foreign-born Filipinas.[17] The difference in main findings may be because of the choice in

TABLE 3—Age-Adjusted Incidence Rates of Breast Cancer Among Women, by Race/Ethnicity, Immigrant Status, and Age Group: California Cancer Registry, 1988–2004

	Aged ≤ 44 Years			Aged 45–54 Years			Aged ≥ 55 Years		
Race/ ethnicity	No. of Cases	Population, No.	Incidence Rate (95% CI)	No. of Cases	Population, No.	Incidence Rate (95% CI)	No. of Cases	Population, No.	Incidence Rate (95% CI)
Non-Hispanic Whites	25 461	84 723 489	27.1 (26.8, 27.5)	46 384	19 287 739	240.7 (238.5, 242.9)	176 576	38 852 016	449.2 (447.1, 451.3)
US-Born Asian									
Aggregated[a]	1 008	7 119 319	36.9 (34.6, 39.3)	1 106	424 901	260.7 (245.5, 276.5)	2 406	827 704	287.6 (276.1, 299.5)
Chinese	343	2 033 578	39.8 (35.6, 44.2)	366	132 745	276.9 (249.1, 306.8)	533	193 280	275.6 (252.4, 300.3)
Japanese	276	1 085 402	23.9 (21.1, 26.9)	439	213 378	205.8 (187.0, 226.0)	1 583	530 587	294.2 (279.5, 309.4)
Filipina	222	2 097 636	43.1 (37.3, 49.4)	208	62 297	334.3 (290.4, 383.0)	207	71 832	263.8 (226.0, 307.3)
Korean	38	...	...	30	...	...	24	...	...
South Asian[b]	53	...	...	44	...	...	44	...	...
Vietnamese	76	...	...	19	...	...	15	...	...
Foreign-Born Asian									
Aggregated[a]	3 450	11 525 032	20.6 (19.9, 21.3)	5 009	3 112 504	161.2 (156.8, 165.8)	8 168	4 228 121	192.6 (188.4, 196.9)
Chinese	965	3 326 882	18.9 (17.7, 20.2)	1 266	936 632	134.8 (127.5, 142.4)	2 259	1 362 745	167.9 (161.0, 175.0)
Japanese	230	663 258	24.8 (21.7, 28.4)	361	183 894	196.0 (176.3, 217.3)	999	389 590	283.6 (264.3, 303.9)
Filipina	1 299	3 177 600	25.9 (24.5, 27.5)	2 181	1 016 353	215.1 (206.2, 224.3)	3 466	1 383 784	245.0 (236.7, 253.5)
Korean	322	1 445 851	15.8 (14.1, 17.8)	426	386 120	110.5 (100.2, 121.5)	464	474 748	90.5 (82.2, 99.4)
South Asian[b]	252	1 141 235	17.3 (15.2, 19.8)	281	221 870	127.2 (112.8, 143.0)	456	230 432	197.1 (177.9, 218.0)
Vietnamese	382	1 770 206	17.4 (15.7, 19.5)	494	367 635	135.2 (123.5, 147.7)	524	386 822	128.3 (117.2, 140.3)

Note. CI = confidence interval. Ellipses indicate that sample sizes were too small to report because of the potential for large errors associated with them. Breast cancer incidence rates are per 100 000 person-years, age-standardized to the 2000 US population.
[a]Consists of the 6 Asian populations: Chinese, Japanese, Filipina, Korean, South Asian, and Vietnamese women.
[b]"South Asian" populations include Asian Indians, Pakistanis, Sri Lankans, and Bangladeshis.

this prior study to randomly distribute immigrant status among the 13% to 22% of cancer cases with unknown registry birthplace data, which resulted in underestimated rates among US-born and overestimated rates among foreign-born women.[22] Second, whereas the 1973–1986 rates were not directly comparable to ours given different populations used for age standardization, the rates among US-born Filipinas were substantially lower in the earlier analysis[17] than they were in our study. The higher rates observed in our study may suggest a large increase in incidence rates in US-born Filipinas, as indicated by the 4% increase per year during the period 1988–2004, as well as immigrant cohort differences in risk factor profiles.

Limitations

Our study had several possible limitations. Our imputation of immigrant status based on SSN, although an improvement over prior methods,[50] had an 84% sensitivity and 80% specificity of classification. The remaining misclassification had a greater relative impact on the smaller immigrant group, usually US-born. The assignment of immigrant status in some groups is fairly sensitive to the SSN age cutoff for imputing immigrant status; for example, using an age cutoff of 20 years instead of 25 years would have yielded 21%, 8%, and 34% fewer US-born Chinese, Japanese, and Filipina breast cancer cases but would have been subject to higher misclassification rates than the cutoff we used. However, the primary function of immigrant status in our study was to indicate acculturation, with those characterized as US-born considered to be more acculturated than those characterized as foreign-born. Even if imputed immigrant status was not completely accurate, it was likely to be correlated with acculturation level.

Our population estimates for Asian populations by immigrant status may also have been subject to error, particularly for specific age groups, which may have biased overall or age-specific rates. Although the Census data we used were the most definitive for estimating populations, counts stratified by immigrant status were available only for a population sample, which reduced their robustness. To evaluate the accuracy of the assumptions underlying our methods for estimating annual populations, we compared our 2004 population estimates to those from the 2005 American Community Survey (ACS), a 2.3% stratified sample of the population,[51] for California. We found less than a 2% difference in the estimates for US- and foreign-born Filipinas, suggesting that our findings for Filipinas were unlikely to be biased by inaccuracies in the denominator data. For Chinese women, our population estimates were higher than the ACS estimates by about 8% for US-born Chinese and about 1.5% for foreign-born Chinese; however, if the ACS estimates were more accurate, the bias was conservative, as we would have underestimated the relative difference in rates between US- and foreign-born Chinese women. Our estimates for US- and foreign-born Japanese women were both 15% higher than the ACS estimates; however, even if the ACS data were more accurate, this difference would be unlikely to explain our finding of a lack of difference in breast cancer rates between US- and foreign-born Japanese women, given that the direction and degree of difference were similar.

Conclusions

Findings from our study support the notion that immigrant status is an important variable for assessing patterns and trends in breast cancer incidence in dynamic populations, like that of California, with significant immigration. It is likely that this measure of acculturation reflects population distributions of nongenetic breast cancer risk factors such as diet, age at menarche, age at first birth, breastfeeding, and other lifestyle and reproductive factors. Future studies should examine the prevalence of established risk factors by immigrant status and socioeconomic status among US Asian populations to understand further how these and other environmental factors explain observed differences in breast cancer incidence rates between US- and foreign-born Asian women.[52] Of particular interest to prevention among US-born Asian women of postmenopausal age will be the prevalence of use of combined hormone therapy and 2 or more alcoholic beverages per day. Further exploration of recent age-specific changes in known and suspected breast cancer risk factors may provide valuable clues to as-yet undiscovered causes of breast cancer (e.g. adolescent diet).

Our findings challenge the current notion based on aggregated data that breast cancer rates are uniformly low among US Asian women by identifying apparent disparities in this diverse population group.[53] Specifically, using novel and robust estimating methods to characterize immigration status, we found that breast cancer rates were high and increasing among US-born Chinese and Filipina women, to a point where contemporary rates may equal or exceed those of non-Hispanic Whites. We also estimated higher rates among younger US-born Chinese and Filipina women than among non-Hispanic White women of comparable age, and a sustained increase in breast cancer incidence rates over time, particularly among US-born Filipina and foreign-born Korean women. Together, our findings demonstrate the need to consider cancer control and policy efforts targeted specifically to these groups, especially among US-born Asian women, and to educate clinicians and women as to the possibility of these previously undocumented risks, especially because the number of US-born Asians is projected to rise in the future. Our data suggest that, within this generation, some groups of Asians are experiencing unprecedentedly high rates of breast cancer. This situation must be further evaluated but also addressed now so that ongoing and future efforts to reduce the breast cancer burden do not overlook Asian women, particularly those becoming more acculturated to the Western lifestyle. Work targeted to these groups should include concerted efforts, such as those with Asian community organizations, to educate young Asian women about lifestyle modifications that can help them reduce their risk of developing breast cancer in the future.

Health disparities and inequities experienced by Asian Americans largely have been underdescribed because of stereotypes about positive health profiles, together with the lack of detailed data about the heterogeneous subpopulations. Examining data enhanced to enable study of cancer incidence in US immigrant groups is an important step toward building an evidence base that can inform health policies for US Asian populations. ■

About the Authors

S.L. Gomez, T. Quach, P.L. Horn-Ross, J.T. Pham, E.T. Chang, T.H.M. Keegan, S.L. Glaser, and C.A. Clarke are with the Northern California Cancer Center, Fremont, CA. M. Cockburn is with the Keck School of Medicine, University of Southern California, CA.

Correspondence should be sent to Scarlett Lin Gomez, Northern California Cancer Center, 2201 Walnut Avenue, Suite 300, Fremont, CA 94538 (e-mail: scarlett@nccc.org). Reprints can be ordered at http://www.ajph.org by clicking the "Reprints/ Eprints" link.

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Contributors

S.L. Gomez supervised the collection of data, originated the study, and supervised all aspects of its implementation. T. Quach conducted the data analysis and led the writing. J.T. Pham and M. Cockburn assisted with the study. All authors helped to conceptualize ideas, interpret findings, and review drafts of the article. All authors certify that they have contributed substantially to the article and approve the final version.

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Human Participant Protection

The Northern California Cancer Center institutional review board approved this study.

References

1. Ries LAG, Melbert D, Krapcho M, et al. SEER Cancer Statistics Review, 1975-2005. Bethesda, MD; 2008. Available at: http://seer.cancer.gov/csr/1975_2005. Accessed January 20, 2009.

2. Miller BA, Chu KC, Hankey BF, Ries LA. Cancer incidence and mortality patterns among specific Asian and Pacific Islander populations in the U.S. *Cancer Causes Control.* 2008;19:227–256.

3. Prehn A, Lin S, Clarke C, et al. *Cancer Incidence in Chinese, Japanese and Filipinos in the U.S. and Asia 1988–1992.* Union City, CA: Northern California Cancer Center; 1999.

4. Gomez SL, Tan S, Keegan TH, Clarke CA. Disparities in mammographic screening for Asian women in California: a cross-sectional analysis to identify meaningful groups for targeted intervention. *BMC Cancer.* 2007;4:201.

5. Wu TY, West B, Chen YW, Hergert C. Health beliefs and practices related to breast cancer screening in Filipino, Chinese and Asian-Indian women. *Cancer Detect Prev.* 2006;30:58–66.

6. Frisbie WP, Cho Y, Hummer RA. Immigration and the health of Asian and Pacific Islander adults in the United States. *Am J Epidemiol.* 2001;153(4):372–380.

7. Gomez SL, Kelsey JL, Glaser SL, Lee MM, Sidney S. Immigration and acculturation in relation to health and health-related risk factors among specific Asian subgroups in a health maintenance organization. *Am J Public Health.* 2004;94:1977–1984.

8. Klatsky AL, Tekawa IS, Armstrong MA. Cardiovascular risk factors among Asian Americans. *Public Health Rep.* 1996;111:62–64.

9. Klatsky AL, Armstrong MA. Cardiovascular risk factors among Asian Americans living in northern California. *Am J Public Health.* 1991;81:1423–1428.

10. Kuo J, Porter K. *Health Status of Asian American: United States, 1992–94.* Atlanta, GA: National Center for Health Statistics, Centers for Disease Control and Prevention; 1998. Vital and Health Statistics No. 298.

11. Ponce NA, Gatchell M, Brown ER. *Cancer Screening Rates Among Asian Ethnic Subgroups.* Los Angeles, CA: UCLA Center for Health Policy Research Policy Brief; 2003.

12. Carrasquillo O, Carrasquillo AI, Shea S. Health insurance coverage of immigrants living in the United States: differences by citizenship status and country of origin. *Am J Public Health.* 2000;90:917–923.

13. Choe JH, Koepsell TD, Heagerty PJ, Taylor VM. Colorectal cancer among Asians and Pacific Islanders in the U.S.: survival disadvantage for the foreign-born. *Cancer Detect Prev.* 2005; 29(4):361–368.

14. Wu TY, Bancroft J, Guthrie B. An Integrative Review on Breast Cancer Screening Practice and Correlates Among Chinese, Korean, Filipino, and Asian Indian American Women. *Health Care Women Int.* 2005;26: 225–246.

15. GLOBOCAN 2002 database: summary table by cancer. Available at: http://www-dep.iarc.fr/GLOBOCAN/Table2_sel.htm. Accessed September 4, 2008.

16. Ziegler RG, Hoover RN, Pike MC, et al. Migration patterns and breast cancer risk in Asian-American women. *J Natl Cancer Inst.* 1993;85:1819–1827.

17. Stanford JL, Herrinton LJ, Schwartz SM, Weiss NS. Breast cancer incidence in Asian migrants to the United States and their descendants. *Epidemiology.* 1995;6(2): 181–183.

18. Ziegler RG, Hoover RN, Nomura AM, et al. Relative weight, weight change, height, and breast cancer risk in Asian- American women. *J Natl Cancer Inst.* 1996;88 (10):650–660.

19. Kelsey JL, Horn-Ross PL. Breast cancer: magnitude of the problem and descriptive epidemiology. *Epidemiol Rev.* 1993;15:7–16.

20. Nelson NJ. Migrant studies aid the search for factors linked to breast cancer risk. *J Natl Cancer Inst.* 2006; 98:436–438.

21. Gomez SL, Glaser SL. Quality of birthplace information obtained from death certificates for Hispanics, Asians, and Pacific Islanders. *Ethn Dis.* 2004;14:292–295.

22. Gomez SL, Glaser SL, Kelsey JL, Lee MM. Bias in completeness of birthplace data for Asian groups in a population-based cancer registry (United States). *Cancer Causes Control.* 2004;15:243–253.

23. Lin SS, O'Malley CD, Lui SW. Factors associated with missing birthplace information in a population-based cancer registry. *Ethn Dis.* 2001;11:598–605.

24. Barnes JS, Bennett CE. The Asian Population: 2000. Census 2000 Brief. US Census Bureau; 2002. Available at: http://www.census.gov/prod/2002pubs/c2kbr01-16.pdf. Accessed January 20, 2009.

25. Gomez SL, Le GM, West DW, Satariano WA, O'Connor L. Hospital policy and practice regarding the collection of data on race, ethnicity and birthplace. *Am J Public Health.* 2003;93:1685–1688.

26. Block G, Matanoski GM, Seltser RS. A method for estimating year of birth using social security number. *Am J Epidemiol.* 1983;118:377–395.

27. Shimizu H, Ross RK, Bernstein L, Yatani R, Henderson BE, Mack TM. Cancers of the prostate and breast among Japanese and white immigrants in Los Angeles County. *Br J Cancer.* 1991;63:963–966.

28. US Census Bureau. Census 1990 Summary Tape File 3. Available at: http://factfinder.census.gov/servlet/DatasetMainPageServlet?_program=DEC&_tabId=DEC2&_submenuId=datasets_1&_lang=en&_ts=282591892840. Accessed January 26, 2010.

29. US Census Bureau. Census 2000 Summary Tape File 3. Available at: http://factfinder.census.gov/servlet/DatasetMainPageServlet?_program=DEC&_tabId=DEC1&_submenuId=datasets_1&_lang=en&_ts=282592024458. Accessed January 26, 2010.

30. Chang ET, Keegan TH, Gomez SL, et al. The burden of liver cancer in Asians and Pacific Islanders in the Greater San Francisco Bay Area, 1990 through 2004. *Cancer.* 2007;109:2100–2108.

31. Keegan TH, Gomez SL, Clarke CA, Chan JK, Glaser SL. Recent trends in breast cancer incidence among 6 Asian groups in the Greater Bay Area of Northern California. *Int J Cancer.* 2007;120:1324–1329.

32. Gomez SL, Le GM, Miller T, et al. *Cancer Incidence Among Asians in the Greater Bay Area, 1990-2002.* Fremont, CA: Northern California Cancer Center; 2005.

33. *R* [computer program]. Version 2.8. The R Foundation for Statistical Computing R Program. Available at: http://www.r-project.org. Accessed January 7, 2010.

34. *SEER*Stat Software* [computer program]. Version 6.1.4. Bethesda, MD: Surveillance Research Program, National Cancer Institute; 2005.

35. *Joinpoint Regression Program* [computer program]. Version 3.0. Bethesda, MD: National Cancer Institute; 2005.

36. Anderson WF, Rosenberg PS, Menashe I, Mitani A, Pfeiffer RM. Age-related crossover in breast cancer incidence rates between black and white ethnic groups. *J Natl Cancer Inst.* 2008;100:1804–1814.

37. Colditz GA, Baer HJ, Tamimi RM. Breast Cancer. In: Schottenfeld D, Fraumeni JF, eds. *Cancer Epidemiology and Prevention.* 3rd ed. New York, New York: Oxford University Press; 2006:995–1012.

38. Deapen D, Liu L, Perkins C, Bernstein L, Ross RK. Rapidly rising breast cancer incidence rates among Asian-American women. *Int J Cancer.* 2002;99:747–750.

39. McCracken M, Olsen M, Chen MS Jr, et al. Cancer incidence, mortality, and associated risk factors among Asian Americans of Chinese, Filipino, Vietnamese, Korean, and Japanese ethnicities. *CA Cancer J Clin.* 2007; 57:190–205.

40. Pike MC, Kolonel LN, Henderson BE, et al. Breast cancer in a multiethnic cohort in Hawaii and Los Angeles: risk factor-adjusted incidence in Japanese equals and in Hawaiians exceeds that in whites. *Cancer Epidemiol Biomarkers Prev.* 2002;11:795–800.

41. Osaka Prefectural Department of Health, Osaka Medical Association, Osaka Medical Center for Cancer and Cardiovascular Diseases. Cancer Incidence and Medical Care in Osaka in 1994 and the Survival in 1990–1997. *Annual Report of Osaka Cancer Registry No. 59.* Osaka, Japan: Osaka Cancer Registry; 1997.

42. Tung HT, Tsukuma H, Tanaka H, et al. Risk factors for breast cancer in Japan, with special attention to anthropometric measurements and reproductive history. *Jpn J Clin Oncol.* 1999;29:137–146.

43. The Research Group for Population-Based Cancer Registration in Japan. Cancer incidence and incidence rates in Japan in 1998: estimates based on data from 12 population-based cancer registries. *Jpn J Clin Oncol.* 2003;33(5):241–245.

44. Nagata C, Kawakami N, Shimizu H. Trends in the incidence rate and risk factors for breast cancer in Japan. *Breast Cancer Res Treat.* 1997;44:75–82.

45. Nakano E, Ikenoue M, Ishikawa A. A study on family types and the employment status of married women in Japan. *Jinko Mondai Kenkyu.* 1982;Apr(162): 2–59.

46. Westley SB, Mason A. Women are key players in the economies of east and southeast Asia. *Asia Pac Pop Policy.* 1998;1–4.

47. Okazaki Y. Coping with the declining birthrate. *Econ Eye.* 1981;2:20–23.

48. Fujino Y, Mori M, Tamakoshi A, et al. A prospective study of educational background and breast cancer. *Cancer Causes Control.* 2008;19:931–937.

49. Tamakoshi K, Yatsuya H, Wakai K, et al. Impact of menstrual and reproductive factors on breast cancer risk in Japan: results of the JACC study. *Cancer Sci.* 2005; 96:57–62.

50. Emerson SS, Emerson JC. Direct standardization of incidence rates in the presence of incomplete data. *Stat Med.* 1993;12:3–12.

51. US Census Bureau. *American Community Survey (ACS).* Washington, DC: US Census Bureau; 2005.

52. Borrell LN, Castor D, Conway FP, Terry MB. Influence of nativity status on breast cancer risk among US black women. *J Urban Health.* 2006;83:211–220.

53. Chen MS. Cancer health disparities among Asian Americans: what we do and what we need to do. *Cancer.* 2005;104:2895–2902.